

SSI2161

PROCIRCUIT™ VOLTAGE CONTROLLED AMPLIFIER

The SSI2161 is a versatile VCA building block for high-performance audio applications. A single channel provides voltage control of a current-mode input and output for a gain range from +20dB to -100dB, with control provided by a ground-referenced -33mV/dB constant.

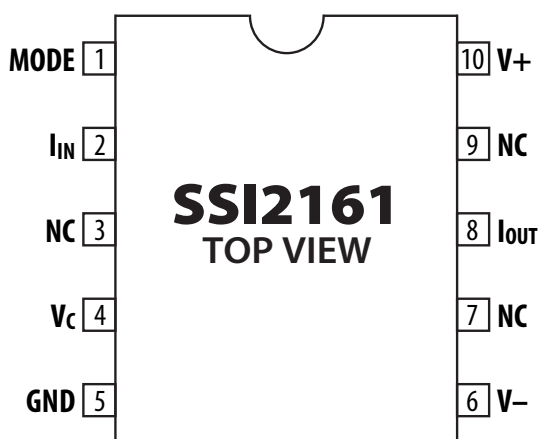
The device offers considerable flexibility for a wide range of design goals and applications. A unique mode control allows selection of Class A, Class AB, or in-between using a single resistor. In addition, improved current handling allows use of lower value input resistors for reduced output noise without loss of headroom. Finally, the VCA can be used as a high-quality OTA building block for a variety of applications such as voltage controlled filters, exponential generators, and antilog converters.

The SSI2161 will operate on supplies as low as +8V for battery-powered devices such as guitar pedals, or up to $\pm 18V$ in systems where maximum headroom is desired.

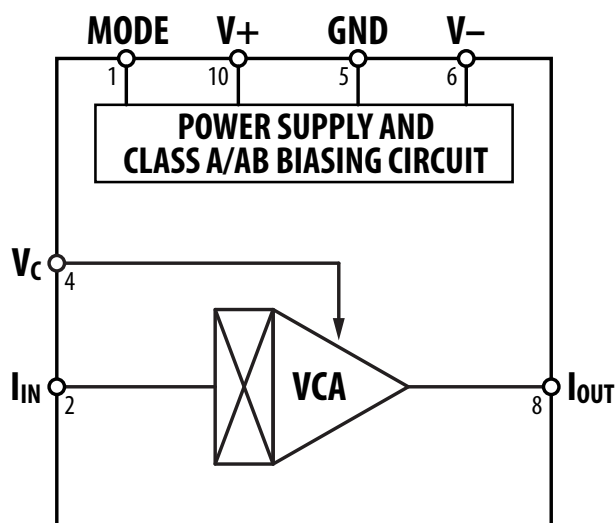
The SSI2161 is part of a family of affordable high-performance VCA's from Sound Semiconductor. The SSI2164 offers four VCA's in a compact SOP package with lowest cost-per VCA, and the SI2162 provides two channels in a small PCB footprint.

FEATURES

- Our Highest Performance VCA in a Compact Package
- Pin-Selectable Class A or AB Operation
- 6dB Lower Noise than SSI2164
- 126dB Dynamic Range (Class AB)
- Low Distortion – Typical 0.025% (Class A)
- Large Gain Range: -100dB to +20dB
- Highly-Compact 10-Lead SSOP Package
- $\pm 4V$ to $\pm 18V$ Operation
- No External Trimming
- Low Control Feedthrough – Typically -60dB



PIN CONNECTIONS
10-LEAD SSOP



**FUNCTIONAL BLOCK
DIAGRAM**

SPECIFICATIONS ($V_S = \pm 15V$, $V_{IN} = 0.775V_{RMS}$, $f = 1kHz$, $A_V = 0dB$, Class AB, $T_A = 25^\circ C$; using Figure 1 circuit without diode)

Parameter	Symbol	Conditions	Min	Typ	Max	Units
POWER SUPPLY						
Supply Voltage Range	V_S		± 4		± 18	V
Supply Current	I_S	Class AB, $V_C = GND$		± 6	± 8	mA
Supply Current	I_S	Class A, $V_C = GND$, $I_M = 1mA$		± 8.0		mA
Power Supply Rejection Ratio	PSRR	60Hz		90		dB
CONTROL PORT						
Input Impedance			2.25	2.5	2.75	k Ω
Gain Constant		After 60 seconds of operation		-33		mV/dB
Gain Constant Temp. Coefficient				-3300		ppm/ $^\circ C$
Control Feedthrough		$A_V = 0dB$ to $-40dB$		-60		dB
Gain Accuracy		$A_V = 0dB$		± 0.30		dB
		$A_V = +20dB$		± 0.55		dB
		$A_V = -20dB$		± 0.55		dB
Maximum Attenuation				-100		dB
Maximum Gain				+20		dB
SIGNAL INPUT						
Input Bias Current	I_B			± 10		nA
Input Current Handling				3.4		mA _P
SIGNAL OUTPUT						
Output Offset Current		$V_{IN} = GND$		± 150		nA
Output Compliance				± 100		mV
PERFORMANCE						
Output Noise		Class AB (20Hz -20kHz, unweighted)				
($I_M = 1mA$)		$R_{IN/OUT} = 7.5k\Omega$		-99		dBu
		$R_{IN/OUT} = 5k\Omega$		-102		dBu
		$R_{IN/OUT} = 3.74k\Omega$		-105		dBu
		$R_{IN/OUT} = 1.8k\Omega$		-107		dBu
		Class A (20Hz -20kHz, unweighted) ¹				
		$R_{IN/OUT} = 7.5k\Omega$		-87		dBu
		$R_{IN/OUT} = 5k\Omega$		-90		dBu
		$R_{IN/OUT} = 3.74k\Omega$		-93		dBu
		$R_{IN/OUT} = 1.8k\Omega$		-99		dBu
Headroom	HR	1% THD		+22		dBu
Total Harmonic Distortion	THD	Class AB (80kHz BW)				
($I_M = 1mA$)		$A_V = 0dB$		0.05		%
		$A_V = 0dB$, $V_{IN} = -17dBu$		0.025		%
		$A_V = +20dB$		0.20		%
		$A_V = -20dB$		0.045		%
		Class A (80kHz BW) ¹				
		$A_V = 0dB$		0.025		%
		$A_V = 0dB$, $V_{IN} = -5dBu$		0.015		%
		$A_V = +20dB$		0.17		%
		$A_V = -20dB$		0.025		%
Unity Gain Bandwidth		$C_F = 10pF$		500		kHz
Slew Rate	SR	$C_F = 10pF$		700		$\mu A/\mu s$

ABSOLUTE MAXIMUM RATINGS

Supply Voltage	$\pm 20V$
Storage Temperature Range	$-65^\circ C$ to $+150^\circ C$
Operating Temperature Range	$-40^\circ C$ to $+85^\circ C$
Lead Temperature (Soldering, 10 sec)	$260^\circ C$
Mode Current (I_M ; Pin 1 to Pin 10 via R_M)	2.0mA
Control Pin Voltage (Pin 4)	V- to V+

ORDERING INFORMATION

Part Number	Package Type/Container	Quantity
SSI2161SS-TU	10-Lead SSOP* - Tube	100
SSI2161SS-RT	10-Lead SSOP* - Tape and Reel	4000

*SSI Package ID "PSSL10"

Mechanical drawing available at www.soundsemiconductor.com

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PIN DESCRIPTIONS

Pin(s)	Name	Description
1	MODE	Current into this pin sets VCA core to operate as Class A (lowest THD), AB (lowest noise), or inbetween, set by external resistor. Leave open for Class AB operation.
2	I _{IN}	Ground-referenced current inputs; each requires RC network.
3	NC	Leave this pin unconnected.
4	V _C	Ground-referenced control port with a –33mV-per-dB constant.
5	GND	Connect to analog signal ground with short, low inductance trace.
6	V–	Negative supply. Recommend 100nF local decoupling capacitor placed as close to package as possible with a low inductance trace to ground.
7	NC	Leave this pin unconnected.
8	I _{OUT}	Ground-referenced current output.
9	NC	Leave this pin unconnected.
10	V+	Positive supply. Recommend 100nF local decoupling capacitor placed as close to package as possible with a low inductance trace to ground.

USING THE SSI2161

The SSI2161 is a single-channel voltage controlled amplifier with a control range from +20dB to –100dB. The VCA is a current-in, current-out device with an exponential voltage control port. Basic operation is described below; see the “Principles of Operation” section for further details on inner workings of the device and an application section that follows.

Signal Input

Figure 1 shows the basic application circuit. A resistor converts the input voltage to an input current, and a 56Ω resistor in series with a 4700pF capacitor connected to ground ensures stable operation. The SSI2161 is quite tolerant of RC network selection, but 56Ω/4700pF has been proven to work well over a wide range of R_{IN} values.

A 3.74kΩ value for R_{IN} is recommended for most applications, but can range from 1.82kΩ to 25kΩ – lower values will produce the best noise performance at some cost in distortion.

Maximum input current handling is approximately 4mA peak. This input current “headroom” is only likely to be a consideration when using R_{IN} values of 2.5kΩ and below with supplies of ±12V and higher. In such cases, one may want to design the signal chain for a maximum input current of 3.6mA to allow adequate headroom.

An optional series-connected 47μF capacitor is recommended for improved control feedthrough.

NOTES:

All resistors shown are ±1% and capacitors 10%

*See text

† Or similar

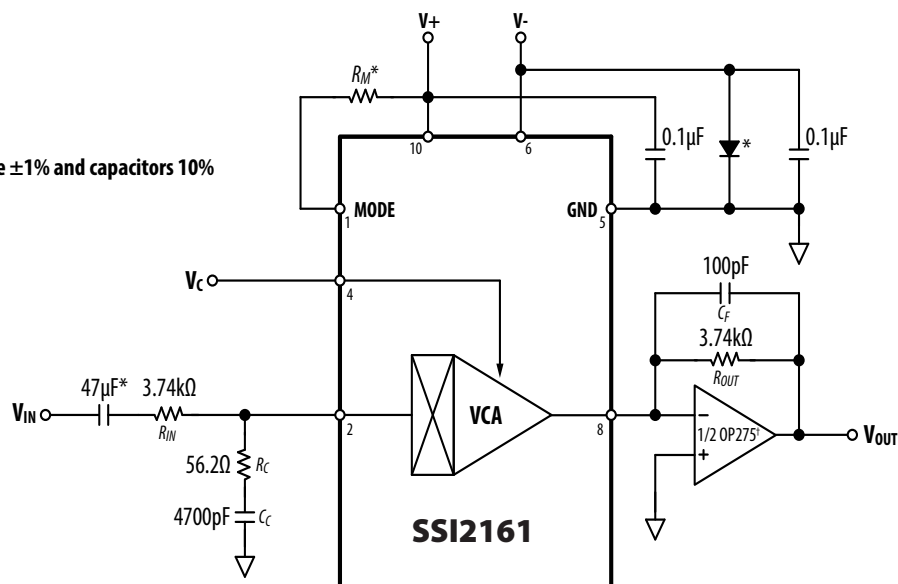


Figure 1: Typical Application Circuit

Signal Output

The output current pin should be maintained at virtual ground using an external amplifier such as a TL072; feedback shown results in unity gain. Many op-amps require a feedback capacitor to preserve phase margin. A value of 100pF will suffice in most cases; larger values can be used to reduce high-frequency noise at the expense of bandwidth. In most cases, headroom will be limited by the op amp – increased headroom can be achieved by using a rail-to-rail amplifier or operating on separate supplies.

The direction of current out of the SSI2161 mimics that of a resistor: current flowing into the input then flows out of the output, and vice versa. This greatly simplifies overall design in filter applications. For audio level applications where the VCA is used only to vary the signal level (e.g., mixer automation, synthesizer VCA module, etc) the combination of SSI2161 VCA and opamp (Figure 1) produces an output whose phase is inverted.

Control Port

The SSI2161 provides exponential control with gain constant of -33mV/dB; to realize the full gain range of +20dB to -100dB, control voltage should span from -660mV to 3.3V, respectively. If only attenuation is desired, the control port can be driven directly from a low impedance voltage-output DAC.

The control input has a nominal impedance of 2.5kΩ, with an internal 10:1 resistor divider. Because of this, any resistance in series with V_C will attenuate the control signal somewhat. If precise control of gain and attenuation is required, buffering the control voltage is suggested.

The 33mV/dB control voltage law is essentially set by transistor physics, and has the property of being proportional to absolute temperature, or approximately 0.33%/°C. This is low enough to be unimportant in most applications, but can be reduced with external temperature-dependent networks. One example is shown in Figure 2 using an inexpensive NTC thermistor (such as Vishay NTCLE100E3272JB0 or similar).

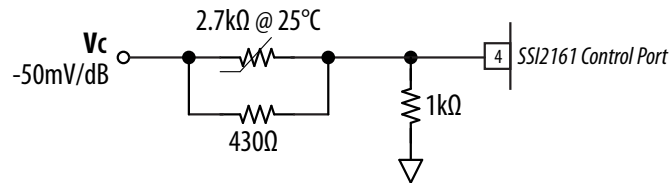


Figure 2: Temperature-Compensated Control Port

Class A Mode Current

The SSI2161's gain core can be biased as Class A, AB, or in-between. Class AB will yield the best noise performance which is achieved with Pin 1 left open. Class A offers lowest distortion and requires connection of resistor R_M between Pin 1 and V_+ .

As Figure 3 shows, mode current I_M affects both noise and distortion. For most applications, a 1mA mode current provides the best overall Class A performance. One can reduce THD further by increasing mode current, but at a significant cost in noise. In addition, increased mode current increases supply current. For example, supply current is typically ± 8.4 mA with a 1.0mA mode current, but jumps to nearly ± 11 mA at 1.5mA. Under no circumstances should mode current exceed 2mA. For some applications, the designer might consider mode current below 1mA for improved Class A noise and power dissipation.

Resistor R_M can be calculated by:

$$R_M = \frac{+V - 0.65V}{I_M}$$

See the "Principles of Operation" section for further discussion on gain core biasing.

Supplies

Supplies from ± 4 V to ± 18 V are possible, which should be regulated and include normal design practices such as bypass capacitors as shown in Figure 1. Since internal protections were added to prevent catastrophic failure that may be experienced during SSM/V2164 asymmetrical power-up, the typical Schottky diode solution is no longer necessary. In most applications, no diode is needed. Modular synthesizer sub-systems may want to include a standard 1N4148-style diode as an extra measure of protection since "hot" plugging of modules may be experienced.

Single supply operation is described in the Applications section.

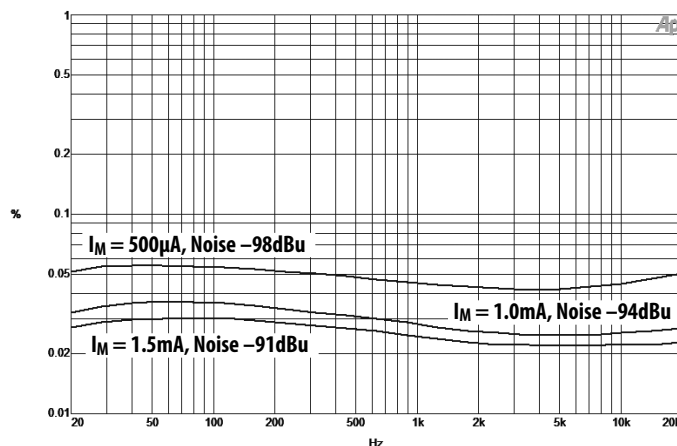


Figure 3: SSI2161 Class A THD+N At Varying Mode Currents
 $R_{IN} = 5k\Omega$, $V_{IN} = 0dBu$, $A_V = 0$, $V_S = \pm 15V$, 22Hz - 80kHz Filter

PRINCIPLES OF OPERATION

VCA Core

The simplified schematic in Figure 4 shows the basic structure of a VCA cell. The gain core is comprised of matched differential pairs Q1 – Q4 and current mirrors Q5, Q6 and Q7, Q8. The current input pin, I_{IN} , is connected to the collectors of Q1 and Q7 and the difference in current between these two transistors is equivalent to I_{IN} . For example, if 100 μA is flowing into the input, Q1's collector current will be 100 μA higher than Q7's collector current.

The control voltage V_C steers the signal current from one side of each differential pair to the other, resulting in either gain or attenuation. For example, a positive voltage on V_C steers more current through Q1 and Q4 and decreases the current in Q2 and Q3. The current output pin, I_{OUT} , is connected to the collector of Q3 and the current mirror (Q6) from Q2. With less current flowing through these two transistors, less current is available at the output. Thus, a positive V_C attenuates the input and a negative V_C amplifies the input. The VCA has unity gain for a control voltage of 0.0 V where the signal current is divided equally between the gain core differential pairs.

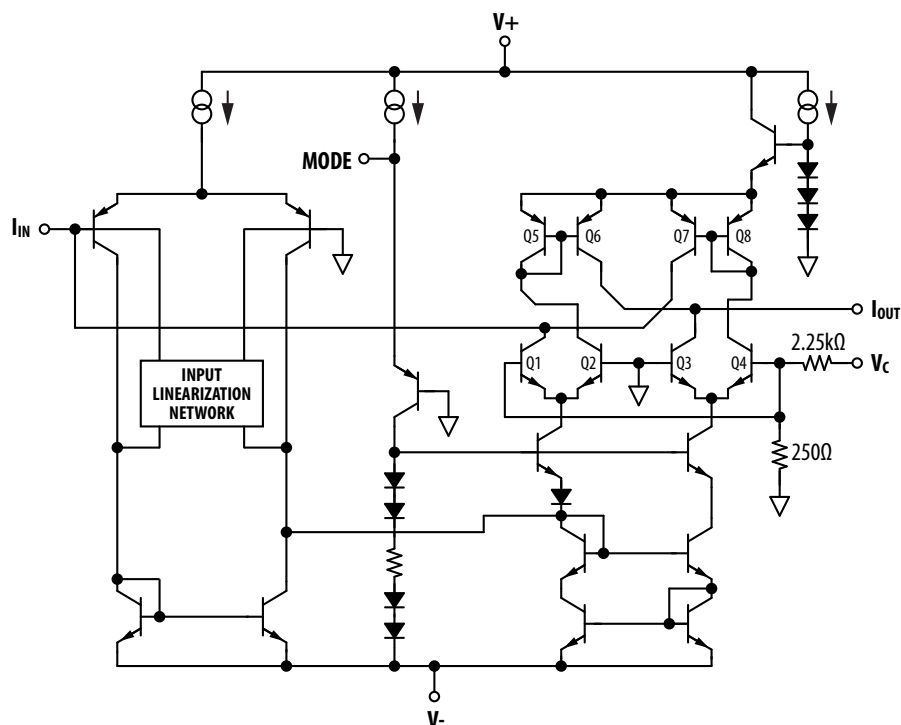


Figure 4: SSI2161 Simplified Schematic

Biasing the VCA Core

VCA's operate by modulating differential currents in a transistor core, a fraction of which is steered to the output in a value set by the control voltage. Such VCA's are generally classified as Class A, Class B and Class AB; terms borrowed from radio transmitter jargon.

In Class A operation the quiescent current in the transistor core is designed to be greater than the maximum input current, so all the transistors remain active at all signal levels. This type of operation produces the lowest distortion, but the high quiescent current level has a severe impact on noise floor and control feedthrough rejection.

In Class B operation the core transistor current is zero and only half the transistors conduct signal at any point of time. Such operation would yield the lowest possible noise floor and near perfect control rejection, but is unfortunately impractical in practice. This is because the transistors effectively disconnect feedback inside the VCA during zero-crossings and low signal levels, potentially causing latch-up or instability. The solution is to arrange the circuit to operate in class A at low signal levels and enter Class B at larger ones. This is known as Class AB operation.

In some applications, distortion may be more important than noise or control feedthrough, in which case it is desirable to raise the point at which the transition from class A to class B takes place. This can be affected by injecting current into the mode pin. The relationship between the input current transition point and the current injected into the mode pin is intentionally non-linear. Figure 5 shows the relationship between the two.

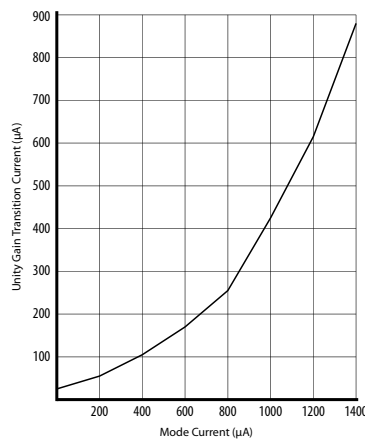


Figure 5: Class A Mode Current to Transition Current Association

For example, supposing the desired transition point is at a peak input voltage of one volt. With a 3.74kΩ input resistor this would correspond to an input current of 200µA, and extrapolating from Figure 5 would require a current of about 650µA to be injected into the mode pin. The mode pin is biased about 0.65V above ground, so a resistor placed between this pin and V+ would see a voltage of 14.35V with a +15V supply. Therefore a resistor value of 22kΩ would work well.

APPLICATION INFORMATION

Please refer to the SSI2164 data sheet and AN701 "Designing Voltage Controlled Filters for Synthesizers with the SSI2164" for comprehensive information that is directly applicable to the SSI2161. Be careful to adjust $R_{IN/OUT}$ as well as the R_C and C_C compensation network to recommended SSI2161 values.

SINGLE SUPPLY OPERATION

By referencing to a pseudo-ground point midway between V+ and V-, the SSI2161 can operate from a single supply between +8V and +36V. An op amp provides a low-impedance reference, from which all ground connections are made.

As shown in Figure 6, a voltage divider comprised of two 10kΩ resistors connected to the non-inverting input provides a ground voltage reference, the output of which is connected to ground on the SSI2161. The SSI2161's input can be referenced to the same ground, or AC coupled as shown in Figure 6. The 10Ω resistor and 22µF capacitor filter noise that may otherwise be present in the pseudo-ground. The control port provides unity gain when V_c is equal to the ground reference voltage, or 4.5V in the case of Figure 6.

To bias SSI2161 gain core as class A, the mode resistor value calculation is modified slightly to:

$$R_M = \frac{\frac{+V}{2} - 0.65V}{I_M}$$

For example, if using 9V supplies R_M should be 3.9kΩ.

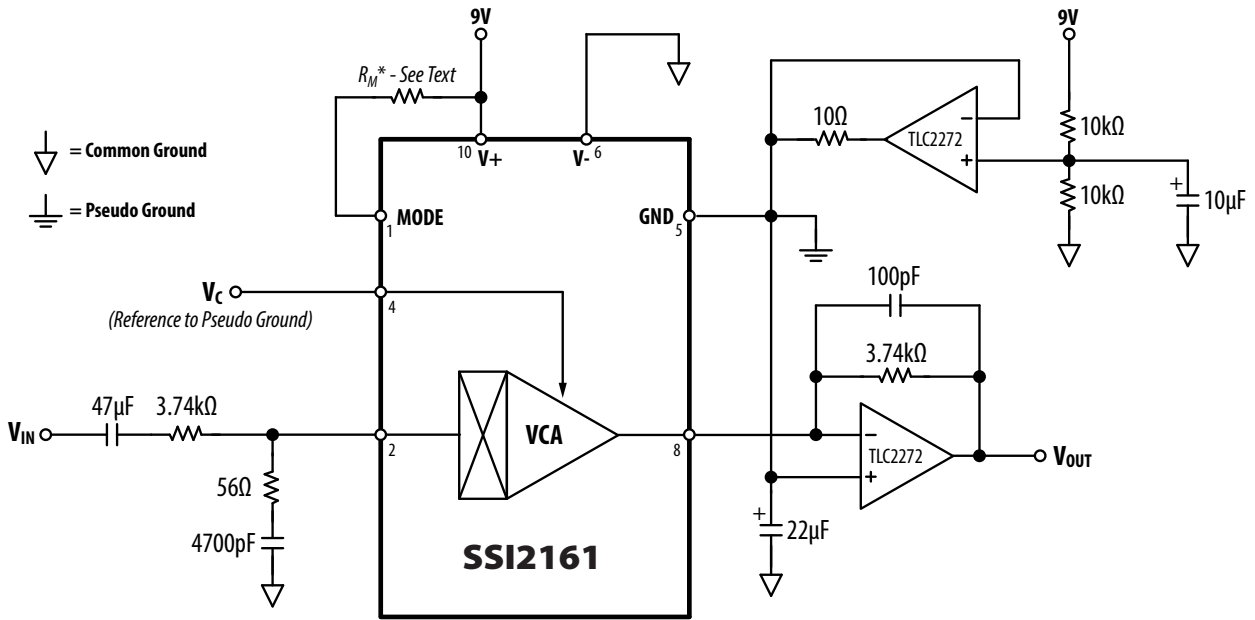
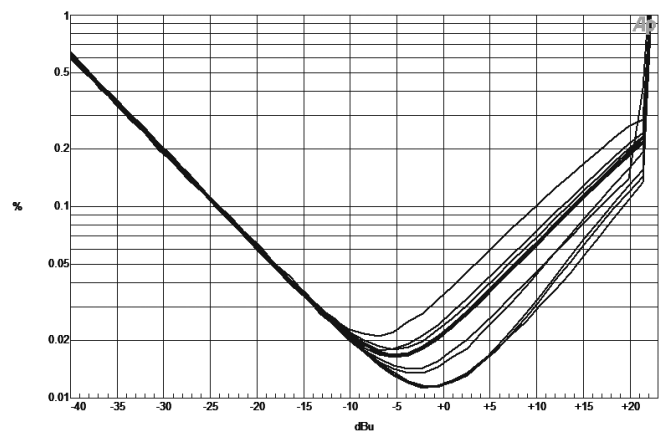
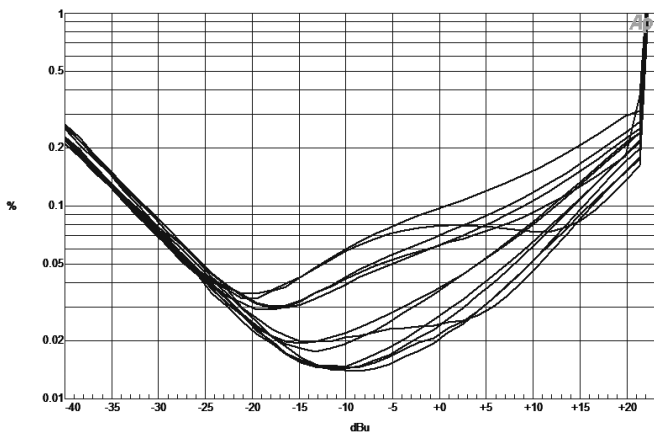
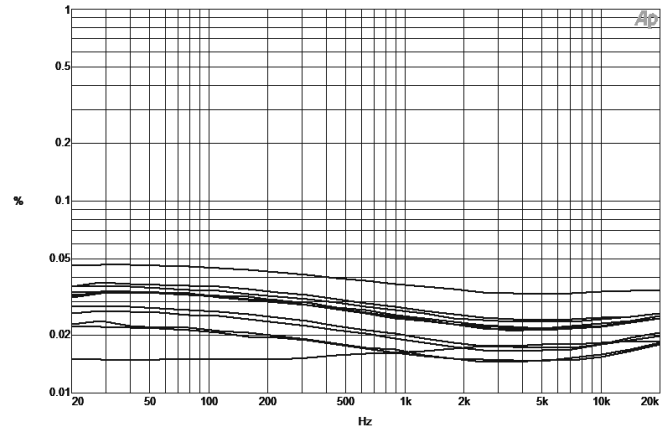
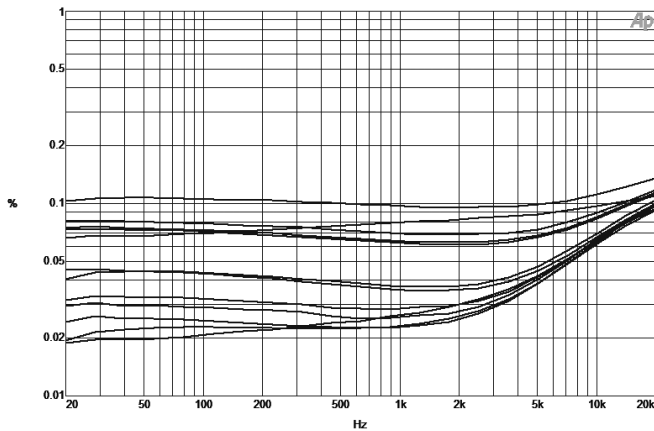
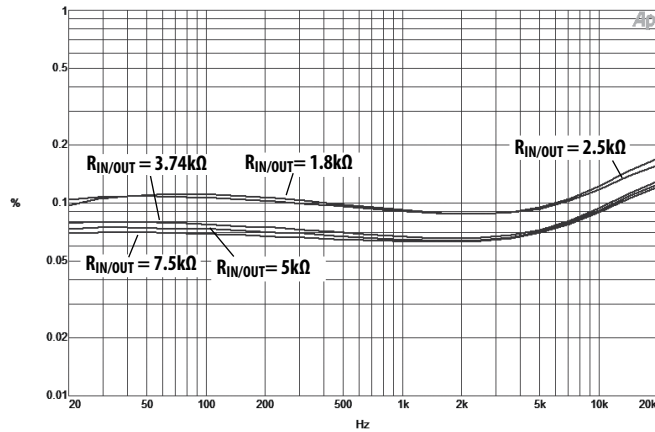


Figure 6: Single-Supply Operation

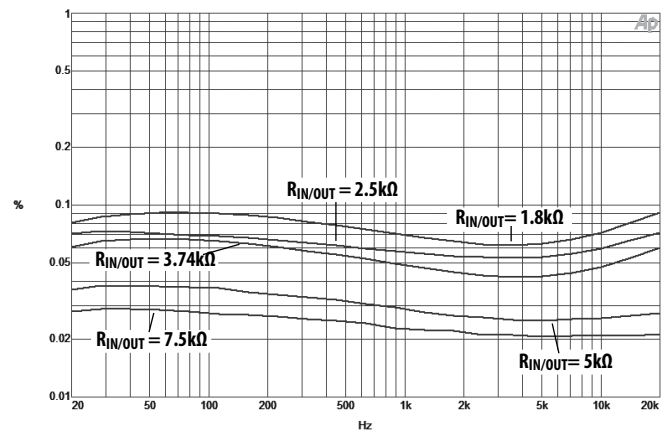
TYPICAL PERFORMANCE GRAPHS*

Figure 1 Application Circuit at $V_S = \pm 15V$, $A_V = 0dB$, $V_{IN} = 0dBu$, $R_{IN/OUT} = 3.74k\Omega$, $f = 1kHz$; unless otherwise noted

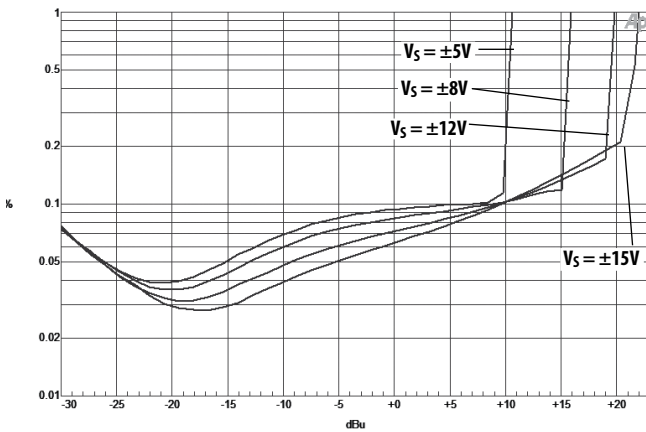




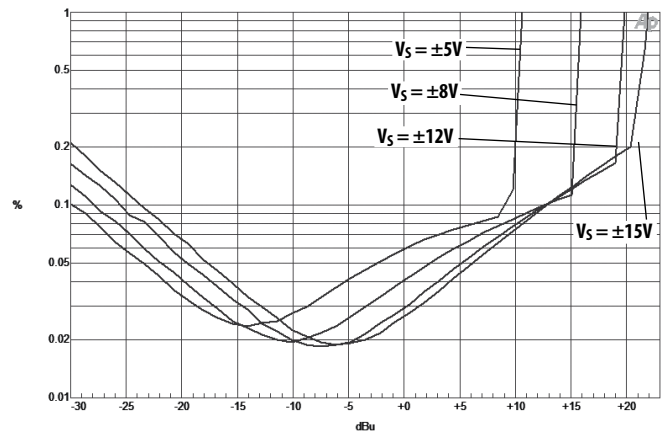
THD+N vs. Frequency vs. $R_{IN/OUT}$
Class AB, 22Hz - 80kHz Filter



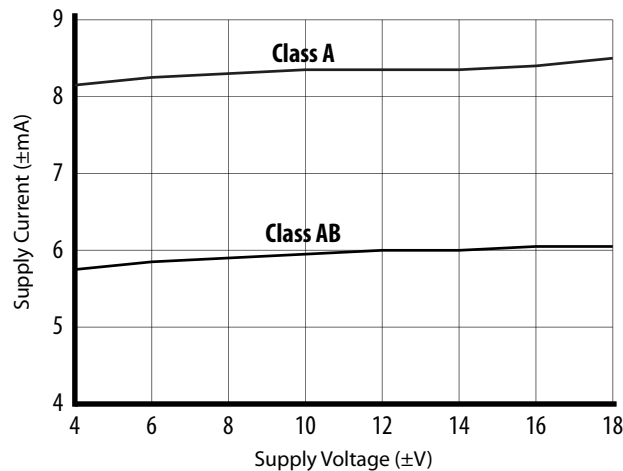
THD+N vs. Frequency vs. $R_{IN/OUT}$
Class A, 22Hz - 80kHz Filter



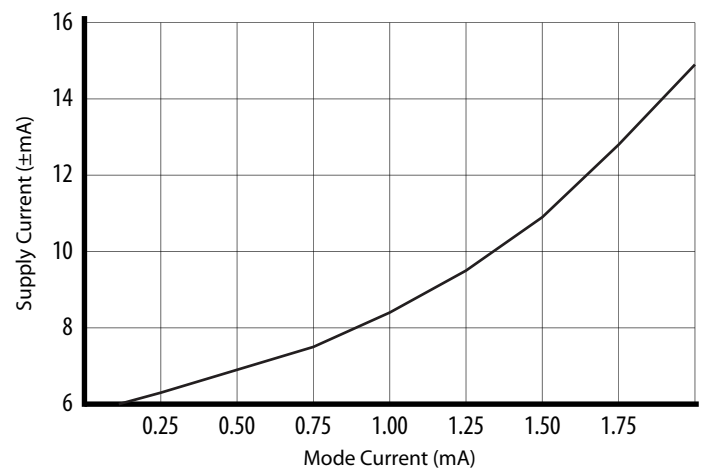
THD+N vs. Amplitude vs. Supply Voltage
Class AB, <10Hz - 22kHz Filter



THD+N vs. Amplitude vs. Supply Voltage
Class A, <10Hz - 22kHz Filter



Supply Current vs. Supply Voltage



Supply Current vs. Mode Current

***About THD+Noise data.** As the name implies, THD+N measures total harmonic distortion *and* noise. In all cases, THD *without* noise will be lower than shown. The noise component will increasingly dominate graph data as signal levels decrease, for example, in THD+N vs. Amplitude graphs. Similarly, an otherwise “apples to apples” comparison between two lines under different noise conditions such as Class A vs. Class AB or differing R_{IN} values may be affected. While one might dismiss the value of THD+N noise measurements, recall that both distortion and noise are undesirable so such information therefore shows all the things you *don't* want, which may be very useful when setting design limits.